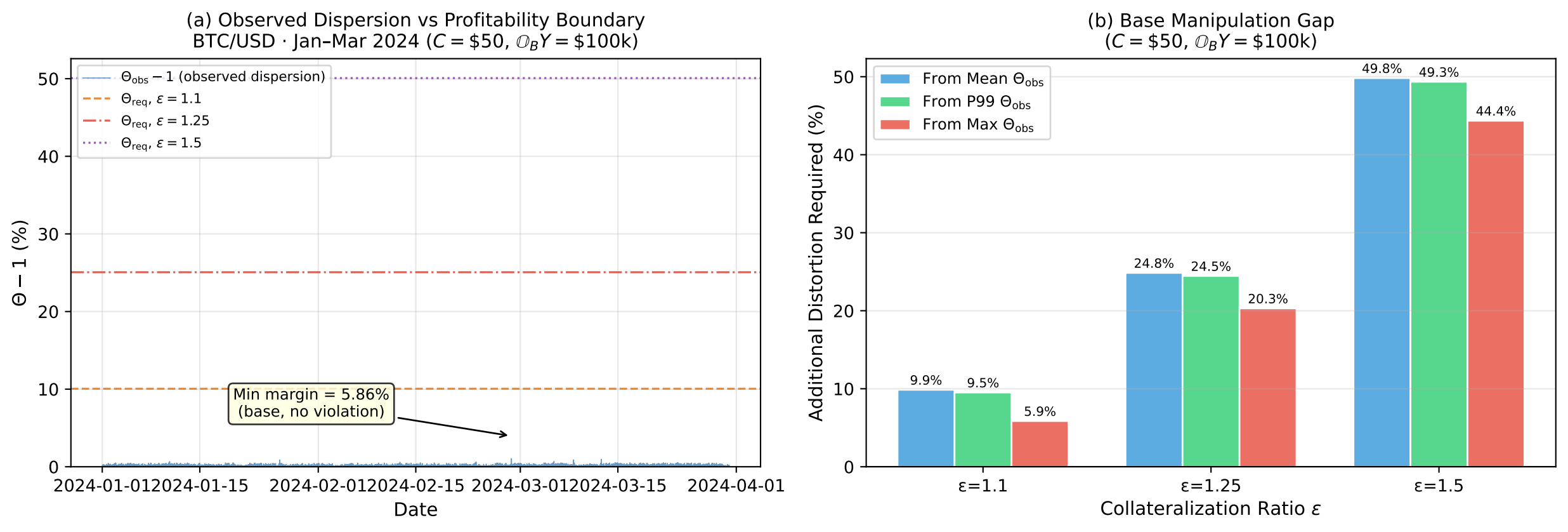

(a) Observed Dispersion vs Profitability Boundary
BTC/USD · Jan–Mar 2024 ($C = \$50$, $\mathcal{O}_B Y = \$100$k)
$\Theta_{obs} - 1$ (observed dispersion)
Θ_{req}, $\varepsilon = 1.1$
Θ_{req}, $\varepsilon = 1.25$
Θ_{req}, $\varepsilon = 1.5$
Min margin = 5.86%
(base, no violation)
$\Theta - 1$ (%)
Date
2024-01-01
2024-01-15
2024-02-01
2024-02-15
2024-03-01
2024-03-15
2024-04-01
(b) Base Manipulation Gap
($C = \$50$, $\mathcal{O}_B Y = \$100$k)
From Mean Θ_{obs}
From P99 Θ_{obs}
From Max Θ_{obs}
Additional Distortion Required (%)
9.9%
9.5%
5.9%
24.8%
24.5%
20.3%
49.8%
49.3%
44.4%
ε=1.1
ε=1.25
ε=1.5
Collateralization Ratio ε